METRIC
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SUPERSEDING
MIL-C-62499(AT)
1 April 1985

COMMERCIAL ITEM DESCRIPTION

CUSHION ASSEMBLIES (METRIC)

The General Services Administration has authorized the use of this commercial item description (CID) as a replacement for MIL-C-62499(AT) which is canceled.

1.0 Abstract. This commercial. item description (CID) covers cushion assemblies used as seats, back rest, and head rests in military tracked vehicles.

2.0 Salient characteristics.

2.1 Material. Cushion assemblies shall not contain materials which would cause discomfort or adversely affect the health of personnel. The use of recovered material made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see 5.4).

2.1.1 Padding. The padding shall consist of a foam-in-place cellular elastomeric material conforming to type V, grade HRII of L-P-386.

2.2 Design and construction. The cushion assembly shall conform to the applicable drawing (see 5.2).

2.3 Physical properties.

2.3.1 Padding.

2.3.1.1 Tear resistance. The padding shall have a tear resistance of not less than 175 Newton per meter (N/m) when tested per ASTM D3574.

2.3.1.2 Adhesion strength. The adhesive bonding material shall have an adhesion strength of at least 137 kilopascals (kPa) when tested per ASTM D429, method B.

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by 1etter to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000.

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2.3.2 Cover.

2.3.2.1 Bursting strength (wrap and filler). The bursting strength of the cover material shall be at least 1100 kPa when tested per ASTM D751.

2.3.2.2 Tear strength (wrap and filler). The tear strength of the cover material shall be at least 27 kilograms when tested per ASTM D2261.

2.3.2.3 Coating adhesion strength. The coating of the cover material shall have an adhesion strength of at least 175l N/m when tested per ASTM D751.

2.3.2.4 Cover material weight. The cover material shell have a total weight of 1247 grams per .836 square meter with the fabric portion being 170 g/m^2.

2.3.2.5 Cover material gage. The total thickness of the cover material shall be 1879 ± l78 micrometers with the skin thickness being 762 ± 25.4 micrometers.

2.3.2.6 Cold impact. The cover material shall not exhibit a failure at -40°C when tested per ASTM D1790.

2.3.2.7 Wear resistance. The cover material shall show no evidence of wearing through when using method 5306 of FED-STD-191 with the following specifics:

a. Load: 1000 grams
b. Wheel type: H-18
c. Revolution: 4000

2.3.2.8 Crocking. Cover material shall show no evidence of crooking when using method 5651 of FED-STD-l91 with a repetition of 10 double strokes.

2.3.2.9 Volatility. The cover material shall have no more than a 6 percent weight loss after being subjected to a heat soak at 93°C far 48 hours, then stabilized at room temperature.

2.3.2.10 Grain retention. A sample of cover material (13 cm x 13 cm) shall show no evidence of loss of grain after being heat soaked at 135° for 1 hour.

2.3.2.11 Color. The color of the cushion assembly shall be black.

2.3.2.12 Color fastness. The cover material shall have a rating of at least “good” with no surface degradation per FED-STD-191, method 5660.

2.3.3 Cushion assembly.

2.3.3.1 Flame resistance. The cushion assembly shall meet the requirements of Federal Motor Vehicle Safety Standard (FMVSS) No. 302.

2.3.3.2 Load capacity. The cushion assembly shall retain at least 90 percent of its original dimensions after a weight of 159 kilograms is applied on a distribution plate with a base diameter of 279.4 ± 25.4 mm for 24 hours. After 24 hours, remove the test weight and plat and, after 15 minutes, measure and examine the cushion assembly for retention of original shape and dimensions.

2.3.3.3 Impact resistance. The cushion assembly shall show no evidence of flattening, tearing or deforming when subjected to 25 000 impacts of 77 ± 2.7 joules using a striker (203 ± 25.4 mm diameter) with a radius of curvature of 406 mm. Apply impacts at the rate of 20 per minute through a swivel joint so that the load is applied over the striker diameter regardless of the angle of the top surface of the test specimen.

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2.4 Environmental. Cushion assembly shall resist atmospheric conditions as specified herein.

2.4.1 High temperature compression set. The cushion assembly shall be heat soaked at 60°C for 47 hours, then the test weight of 159 Kg shall be applied for 1 hour at the 60°C temperature environment. Remove the test weight, allow the cushion assembly to stabilize at room temperature, then measure the thickness and calculate the compression set. The cushion assembly shall not exceed a compression set of 20 percent. (Note: The compression set is the difference between the original and final thicknesses, divided by the original thickness.)

2.4.2 Low temperature compression set. For low temperature compression set, repeat test of 2.3.1 except the temperature shall be -56°C and the compression set shall not exceed 25 percent.

2.4.3 High temperature. The cushion assembly shall retain at least 90 percent of its original dimensions when tested in accordance with MIL-STD-810, method 501.2, procedures I and II, with the following specifics:

a. Procedure I: 71°C and 15 percent relative humidity.
b. Procedure II: 60°C for 48 hours and 15 percent relative humidity.

After stabilizing at room temperature, the cushion assembly shall be tested in accordance with 2.2.15.

2.4.4. Low temperature. The cushion assembly shall retain at least 90 percent of its original dimensions when tested in accordance with MIL-STD-810, method 502.2, procedures I and II, at -56°C for 24 hours. After stabilizing at room temperature, the cushion assembly shall be tested in accordance with 2.2.15.

2.4.5 Humidity. The cushion assembly shall, be tested in accordance with MIL-STD-810, method 507.2, procedure II. The conditions shall confirm to figure 507.2-2, cycle 4, except the humidity limits shall be 5 and 100 percent.

2.4.6. Fungus. The cushion assembly shall be constructed of materials that will not support fungus growth.

2.4.7 Submergence. The cushion assembly shall be submerged in water to a depth of 2.5 cm for at least 30 minutes. (NOTE: Submergence depth shall be measured between the water surface and the top surface of the cushion assembly.) After submergence, the cushion assembly shall be allowed to dry for 48 hours, and then tested in accordance with 2.2.15.

2.4.8. Chemicals. Surface of cushion assembly shall not have any deterioration such as, tackiness or loss of color when exposed to fuel, hydraulic fluid, or cleaning agents, and shall meet the requirements of this CID following exposure to chemicals.

2.4.9 Waterjet spray. The cushion assembly shall show no signs of damage when subjected to waterjet spraying through a nozzle having an orifice diameter not larger than 6.3 mm with the pressure not more than 172 kPa at a rate of .836 square meters per minute, receiving a 200 percent coverage, then allowed to dry for 16 hours at room temperature, then load tested (see 2.2.15) to determine if requirements of this CID are met

2.5 Identification and marking. Cushion assemblies shall be identified and marked in accordance with applicab1e drawings and shall include, as a minimum, the manufacturer’s CAGE code and part number (see 5.2).

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2.6 Workmanship. Workmanship shall be in accordance with the manufacture of a high quality product. This shall be evidenced by the absence of defects which could be detrimental to the appearance or serviceability of the cushion assembly. Defects such as: cracks, checks, tears, or the cover material is tacky, sticky, emits odors, or stains clothing shall not be acceptable.

3.0 Quality assurance provisions.

3.1 Responsibility for inspections. The contractor is responsible for the performance of all inspection (examinations and tests).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications standards and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.0 Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order (see 5.2).

5.0 Notes.

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

5.1 Addresses for obtaining copies of referenced documents.

5.1.1 Government specifications and standards. Copies of MIL-STD-810 "Environmental Test Methods and Engineering Guidelines"; FED-STD-191 "Textile Test Methods"; and L-P-386 "Plastic Material, Cellular, Urethane (Flexible), are available from the Navy Publications and Printing Service Office, Standardization Document Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

5.1.2 Non-Government publications. Copies of ASTM D429 "Rubber Property-Adhesion to Rigid Substrates, Test Methods for; ASTM D751 "Coated Fabrics, Methods of Testing"; ASTM D1790 "Brittleness Temperature of Plastic Sheeting by Impact, Test Method for"; ASTM D2261" Standard Test Method for Tearing Strength of Woven Fabrics by the Tongue (Single Rip) Method (Constant-Rate of-Extension Tensile Testing Machine)"; and ASTM D3574" "Flexible Cellular Materials", are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103. Copies of FMVSS No. 302 are available from the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.

5.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this commercial item description.
b. Title, number and date of applicable drawings see 2.2).
c. Part numbers (see 2.5).
d. Applicable level and packaging requirements (see 4.0).
e. Issue of the DODISS to be cited in the solicitation and the specific issue of individual documents referenced in 5.1.

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5.3 Regulatory requirements. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practicable.

5.4 Cross-reference data. Cushion assemblies conforming to this CID are interchangeable/substitutable with cushion assemblies conforming to MIL-C-62499(AT).

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